

CANCER DIAGNOSIS: FINAL REPORT

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ABSTRACT

This project explores the correlation between visual patterns in unstructured MRI image data and clinical subtypes stored as structured data. The core task we will address is patient tumor grade classification. In the patient population used for our project this involves predicting the Nottingham Grade of tumor defined as Grade 1, 2, or 3. By accurately predicting tumor grades, computational models can assist in determining the aggressiveness of the disease, which directly influences treatment planning.

1 [2 POINTS] INTRODUCTION AND PROBLEM DEFINITION (1-1.25 PAGES)

Multimodal deep learning strategies are becoming increasingly popular for the task of predicting Nottingham tumor grade in recent research, but challenges still remain. When using open source datasets, strategies often need to consider the highly imbalanced historical grade data that is available for training. We are using a Deep Multiplex Graph Infomax architecture Park et al. (2020) for this project. We aim to compare new strategies for data processing and model training to the approach discussed in Kim et al. (2023), which highlights the utility of this architecture for testing heterogeneous graph learning to predict tumor grade. Our new strategies concentrate on the performance of Nottingham Grade predictions as they relate to the overall imbalanced distribution of grade common in many patient populations. In our project we introduce new data processing pipeline methodologies and architectural components. We show that these changes can indicate a strong potential for generalizing meaningful predictions across all Nottingham tumor grades.

1.1 THE DATASET AND CORE TASK

The core task we will address is Tumor Grade Classification, which involves predicting the histological grade of a tumor (e.g., Grade 1, 2, or 3). To accomplish this, we utilize the Duke-Breast-Cancer-MRI dataset, published by The Cancer Imaging Archive (TCIA), which focuses on the diagnosis and prognosis of invasive breast cancer. This task addresses the challenge of precise breast cancer characterization by tackling the correlation between visual patterns in MRI scans and genomic/clinical subtypes. By accurately predicting tumor grades, computational models can assist in determining the aggressiveness of the disease, which directly influences treatment planning.

1.2 THE MULTIMODAL NATURE OF BREAST CANCER DIAGNOSIS

Breast cancer diagnosis is inherently multimodal because a single perspective often fails to capture the full pathology. Our approach leverages two distinct modalities to construct a comprehensive patient profile. The primary imaging component of this study utilizes Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI), a modality renowned for its high sensitivity in detecting and characterizing oncological lesions. To complement the high-dimensional radiological features, a tabular modality comprising structured non-image clinical variables are intergrated. The clinical dataset includes patient demographics, clinical, pathology, treatment, outcomes, and genomic data. These features provide a longitudinal and biological context that imaging alone may overlook. Combining the spatial information from MRIs with the systemic biological information from clinical records allows for a holistic patient representation that is critical for accurate grade classification.

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1.3 KEY CHALLENGES

While integrating multiple modalities is theoretically advantageous, our baseline analyses revealed that simply adding image features or graph structure does not automatically produce a better grade-aware representation. We hypothesize that patients who share similar non-image clinical features are likely to manifest similar visual tumor patterns, and that connecting these patients via a graph neural network (GNN) can act as a powerful regularizer.

However, a central challenge observed during baseline testing is achieving balanced multimodal fusion. Our intrinsic evaluations demonstrated that different multimodal architectures often over-emphasize one branch; for instance, some models' variance was overwhelmingly image-dominated, while others were shaped almost entirely by tabular node attributes. Therefore, the primary obstacle of this project is not merely building a larger multimodal model, but rather engineering a consensus representation in which the two modalities contribute in a balanced and clinically meaningful way.

2 RELATED WORK AND BACKGROUND

2.1 RELATED DATASETS

In our project we are using the Duke-Breast-Cancer-MRI (Duke-Breast) dataset, published by The Cancer Imaging Archive (TCIA). This dataset focuses on the diagnosis and prognosis of invasive breast cancer using multi-modal data. The dataset was extensively analyzed and curated in Saha et al. (2018).

Other research that assisted in our decisions about study design is described in the paper "Breast-DCEDL: A Comprehensive Breast Cancer DCE-MRI Dataset and Transformer Implementation for Treatment Response Prediction" Fridman et al. (2025). This paper introduces BreastDCEDL, a multi-center deep learning dataset of 2,070 patients from multiple data sources including Duke-Breast. It explains a strategy based on a Vision Transformer model using RGB-fused temporal phases to predict patient subgroups.

An additional dataset used for similar research is described in the paper "MAMA-MIA: A Large-Scale Multi-Center Breast Cancer DCE-MRI Benchmark Dataset with Expert Segmentations" Garucho et al. (2026). This dataset contains 1,506 annotated DCE-MRI studies from across 25 health centers. The expert-based categorizations used to group patients in this dataset assisted in the decision process used for our domain-knowledge based tests described in section 3.

2.2 UNIMODAL BASELINES

To establish foundational benchmarks before introducing complex multimodal fusion, we analyze two distinct unimodal approaches.

For the visual-only baseline, we utilize ResNet He et al. (2016), a highly effective deep convolutional neural network (CNN). ResNet's architecture is widely recognized for its implementation of "skip connections," a structural innovation that effectively mitigates the vanishing gradient problem and enables the robust training of exceptionally deep networks on medical imagery.

Conversely, to isolate the predictive signal inherent strictly within structured clinical data, we employ XGBoost Chen & Guestrin (2016) as our tabular-only baseline. This supervised, multi-class gradient-boosted tree algorithm processes exclusively non-image patient variables. By training on ordinal-encoded clinical features from a separate set of labeled patients and subsequently evaluating on a held-out cohort, XGBoost provides a lightweight yet robust reference for the diagnostic value of clinical records alone.

2.3 SIMPLE MULTIMODAL BASELINES

We benchmark the performance of two multimodal baselines, each with two modalities - tabular data and image data.

One of the simple multimodal baselines is a 2-layer multilayer perceptron neural network trained on the non-image features and image features. The image data is encoded using the ResNet-18 strategy

described above. For the basic multimodal test, the categorical non-image data is preprocessed only by using one-hot encoding and then concatenated with the encoded image data to create the model input.

The other simple multimodal baseline is a 1-layer MLP with two projection layers prior to MLP that project image embeddings and tabular features to the same dimension. The projection layers are trained during model training. Specifically, for each series of data - pre-contrast, post-contrast, and subtraction, we sample 3 slices around the center of the tumor. We treat each series as a channel in RGB channels and fuse slices from three series into a single RGB composite image. The RGB images are then cropped to 256x256 pixels surrounding the tumor region. We use Swin transformer as the image encoder due to its linear computational complexity with respect to input size and its ability to learn multi-scale feature representation, which is especially useful for medical image analysis where a understanding of both location and semantic meaning is essential. Clinical features are tabular data, and we use one-hot encoding for categorical features that have more than 2 categories, and binary coding for binary categorical features. We use standard scaling on numerical features. Since the image embedding dimension (768-dim) is much larger than feature matrix (50-dim), we use two learnable project layers to project them to the same 64-dim to force the model to learn from both modalities. The two vectors are concatenated and passed through a single-layer MLP for the final classification task. We expect this baseline to capture the local morphological features and the location of the features in the images, the semantic meanings of clinical features, and the complementary information from both data modalities to make an informed prediction.

2.4 PRIOR WORK

One of the studies used as foundational research for this project is “Deep multiplex graph info-max: Attentive multiplex network embedding using global information” Park et al. (2020). This is an unsupervised graph neural network embedding method that integrates learning multiple types of relationships between the same set of graph nodes. It explains a consensus regularization strategy that accomplishes this by using an attention mechanism to prioritize the most important node relationships.

The Kim et al. (2023) study “Heterogeneous Graph Learning for Multi-modal Medical Data Analysis” extends the DMGI research by fusing medical image data with data including non-image clinical characteristics to learn patient similarities. The data processing strategies and model architecture described in this paper were used as the initial baseline for the new strategies that were explored in our project.

Recent advancements highlight the synergy of combining multi-scale and multi-modal features for breast cancer diagnosis. For example, Wang et al. (2026) introduced MuTriM, an attention-based fusion framework integrating longitudinal DCE-MRI radiomics with whole-slide pathology images to predict recurrence and treatment benefits. While their focus includes pathomics, it reinforces the necessity of cross-modal integration to capture tumor heterogeneity. Similarly, hybrid approaches have proven highly effective in subtyping; Shangquan et al. (2026) developed a stacking ensemble model that combines deep features extracted by a 3D ResNet-50 network with engineered radiomic features from DCE-MRI to successfully differentiate between Luminal A and B breast cancer subtypes. Beyond our baseline, other contemporary studies have adapted GNNs to solve complex topological challenges in breast cancer MRI. Jibon et al. (2026) demonstrated that applying Graph Convolutional Networks (GCNs) combined with advanced texture feature extraction (like Dual-Tree Discrete Wavelet Transform) on DCE-MRI yields exceptional classification accuracy by effectively identifying texture patterns across different tissue types. More aligned with our clinical integration goals, Salehzehi et al. (2025) designed a Graph Isomorphism Network (GIN) that constructs 3D volumetric graphs from multiparametric MRI and clinical features to predict HER2 status, TNM stage, and histological grades in a multi-label setting. Furthermore, Ceccarelli et al. (2024) proposed MUGI-MRI (Multiplex Graph Neural Network for Integration of MRI), which models the temporal instants of DCE-MRI sequences as distinct layers in a multiplex network, proving that aggregating temporal graph data significantly improves model sensitivity and specificity over standard instant-wise analysis.

2.5 RELEVANT TECHNIQUES

While the Duke-Breast dataset is popular for multimodal research studies, much of the research does not utilize the entire population of 922 patients in the dataset. For the task of predicting tumor grade, this may be due partly to the missing grade label for 273 patients in the “Nottingham Grade” column of the “Clinical and Other Features” non-image data. Using a strategy described in the study “A deep learning framework to stratify Nottingham histologic grade 2 breast tumors based on dynamic contrast-enhanced MRI” Hadidchi et al. (2026) we derive tumor grade based on using the ‘Tumor Grade(T) (Tubule)’, ‘Tumor Grade(N) (Nuclear)’, and ‘Tumor Grade(M) (Mitotic)’ data that is available for the 273 patients that do not have Nottingham Grade populated. This data was used to calculate a score as defined in the ICCR (International Collaboration on Cancer Reporting) guidelines Ellis et al. (2022). As a validation step, tumor grade was also assigned in this way where it is already available and compared against the existing tumor grade for 649 patients in the dataset to make sure all grades matched correctly.

The original HetMed study Kim et al. (2023) was used as the initial structure for our experiments. Our tests are similar to the original HetMed architecture in the following ways: 1. Our project uses the Deep Multiplex Graph Infomax architecture. 2. We utilize four groups of patient features to define the metapath views used to calculate each patient node embedding. 3. We compare cluster-based feature groups and domain-knowledge based selection of feature groups. 4. The thresholds used to define cosine similarity when constructing the graph adjacency matrix are based on variation within non-image feature groups. We adjusted thresholds for cluster-based feature groups and domain-knowledge based on the strategy for adjusting similarity thresholds described in the original HetMed study. 5. We evaluate results of learned patient representations on the downstream task of grade classification.

Our project tested variations to the original HetMed data processing in addition to architectural changes that are described in Section 3 - Task Setup and Data.

3 [1 POINTS] TASK SETUP AND DATA

We use the following structured image data and unstructured non-image data from the 922 patients in the the Duke-Breast-Cancer-MRI (Duke-Breast) dataset.

- Modality 1 (Vision): Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI). These 3D volumetric scans capture the structural morphology, texture, and vascularity of the tumor.
- Modality 2 (Tabular/Text): Clinical and genomic data. This includes non-image features such as patient demographics (age), treatment and genomic data.

To evaluate clustering on non-image patient features we used Silhouette Score, Within-Cluster Average Distance, Between-Cluster Average Distance, and Davies- Bouldin Index.

For Nottingham Tumor Grade classification we evaluate results using Macro-F1 and Micro F1.

Training, Validation, and Test Datasets: A summary of the data in splits used for training, validation and testing with a comparison to the original dataset is included below.

Split	Total	Grade 1	Grade 2	Grade 3
Original	922	180 (19.52%)	459 (49.78%)	283 (30.69%)
Training	645	150 (23.26%)	297 (46.05%)	198 (30.70%)
Validation	92	10 (10.87%)	54 (58.70%)	28 (30.43%)
Test	185	20 (10.81%)	108 (58.38%)	57 (30.81%)

Table 1: Distribution of data across different splits and grades.

We use the 5 components from the original HetMed architecture described in Section 2 - Relevant Techniques and test key variations to previous strategies in our experiments. These variations are described below.

HetMed + N4 + Grayscale: This strategy differs from the initial architecture described in Section 2 - Relevant Techniques by using a domain-specific image encoder instead of using continued pretraining to create a new image encoder. The data processing in this experiment also utilizes image correction strategies that differ from the original HetMed architecture. In addition, this strategy uses one grayscale image replicated three times but uses only the post-contrast image for the image modality input.

The first version of this experiment was based on using the Densenet-121 encoder. A later version of our tests with this architecture used the Densenet-121-RadImageNet Mei et al. (2022) domain-specific encoder along with modifications to hyperparameters and produced an improvement in results over the first version.

The N4 correction and intensity thresholding methodology that was used is based on research from Mawlood & Alkarawi (2026), Kanakaraj et al. (2024), and Tustison et al. (2010). For each patient, the pre-contrast image was used to define the N4 correction and intensity thresholding that was then applied to the post-contrast image. The final image modality data for each patient was one corrected grayscale post-contrast image that was replicated three times and used as input for the Densenet-121-RadImageNet encoder.

4 [1 POINTS] BASELINES

We tested the following simple multimodal baselines.

1. Simple multimodal baseline 1: A two-layer MLP that combines non-image clinical data with image features. It extracts 512-dimensional embeddings from medical image slices using a pre-trained ResNet18 encoder and processes structured tabular clinical features through categorical encoding. The non-image and image data are concatenated to perform classification of tumor grade.
2. Simple multimodal baseline 2: A single-layer MLP with two learnable projection layers. The image embeddings are generated using Swin Transformer. Batches include images from each class (tumor grade) and are mostly balanced. Early stopping is used to avoid overfitting because we used a subset of samples for baseline testing (100 out of 922 samples). The primary reason for using a subset of samples is compute constraints. One-hot encoding is used to encode categorical features and standard scaling is performed on numerical features. Projection layers are used to ensure that images and clinical features are in the same dimension so that no data modality dominates the learning.

Swin transformer uses shifted window for attention and self-attention is computed only within a window with a linear complexity $O(N)$, which is more efficient than vision transformer with a quadratic complexity $O(N^2)$. In addition, it merges patch layers and builds multi-scale feature representation, which is essential to medical image analysis where both fine-grained and coarse features need to be identified. The tabular features are scaled by standard scaling to match the distribution of the output vector from Swin transformer. In addition, the project layers project image and tabular vectors to the same dimension. These two design choices force the model to learn from both data modalities.

Since it's simple concatenation of two data modalities, the cross-modal interactions are learned only through the downstream MLP layer. With a nonlinear RELU layer, the MLP classifier is able to capture nonlinear interactions between the two modalities. However, concatenation as a fusion strategy is not expressive enough to capture fine-grained cross-modal interactions, as it delegates all inter-modal reasoning to the downstream layers rather than allowing modalities to inform each other during representation learning.

We also tested the following competitive baselines.

1. System 1 is the Gemini family of multimodal foundation models. We evaluate Gemini 2.5 Flash and Gemini 3 Flash Preview as prompt-based competitive multimodal baselines. These models natively support joint image-text reasoning, so our implementation uses them in a zero-shot setting without any task-specific fine-tuning. For each patient, we provide the cropped RGB composite MRI images together with the same structured clinical feature list used elsewhere in our pipeline, and ask the model to directly predict the Nottingham grade.

This setup makes Gemini a strong off-the-shelf multimodal baseline because cross-modal interactions are handled internally by the pretrained model rather than by a separate fusion module designed by us. Using the same inference framework, we also run feature-only and image-only ablations to isolate the contribution of each modality while keeping the prompting and decoding procedure fixed.

2. System 2 uses the MedGemma multimodal model Sellergren et al. (2025) to generate grade predictions based on prompting. For each patient, structured clinical non-image data was converted into formatted values in text and provided to the prompt along with image data. The best performing version of this test included three center slices of RGB image data for each patient created by merging each of the three images across three series from the patient study. These three series were selected based on a defined prioritization for each of the three different slices to consider temporal relationships between different phases of the MRI scan including pre-contrast, post-contrast, and the timeframe between pre-contrast and post-contrast imaging.
3. System 3 is HetMed framework Kim et al. (2023), whose key insight is structuring patient data into multiplex networks to fully leverage the complementary relationships between image data and clinical data. The details can be found in section 3. As a competitive multimodal baseline, resnet18-SimCLR Chen et al. (2020) pretrained with STL10 dataset was employed. Multi-Instance Contrastive Learning (MICLe) Azizi et al. (2021) loss was adopted for the image encoder. Non-image feature split was based on predefined groups and the threshold of multiplex networks were cosine similarity 0.8.
4. System 4 is a vision-language model Ovis2-4BLu et al. (2024). It aims to solve the structural misalignment in vision-language models where the vision embeddings are continuous vectors from MLP but the text embeddings are discrete. It performs better than other VLMs in the same size category. The image processing is the same as described in simple multimodal baselines. We use few-shots prompting to query the model for classification tasks. Specifically, we select a patient from each grade and use them as examples in the prompt. Both processed RGB composites and clinical features are included in the prompt. We then ask the model to predict tumor grade based on RGB composites and clinical features of an unseen patient.

In our baseline tests, the Swin Transformer + MLP architecture demonstrated the best results, outperforming all other multimodal, specialized, and baseline models. While integrating tabular and visual data generally produced better results, the success of the tests using XGBoost showed the potential for structured tabular data to have more predictive power than visual features alone. Among the generative models that were tested, the Gemini 3 Flash Preview shows significant improvements over Gemini 2.5 Flash, though it still trails behind the top-performing hybrid vision-transformer approach. Specialized medical models like MedGemma provide competitive results but failed to surpass the lead held by other multimodal combinations. Ultimately, baseline tests on the data indicated that a strong visual component paired with tabular data integration produced the best results when compared to generalist LLMs and unimodal baselines.

5 [3 POINTS] PROPOSED MODEL (>1 PAGE)

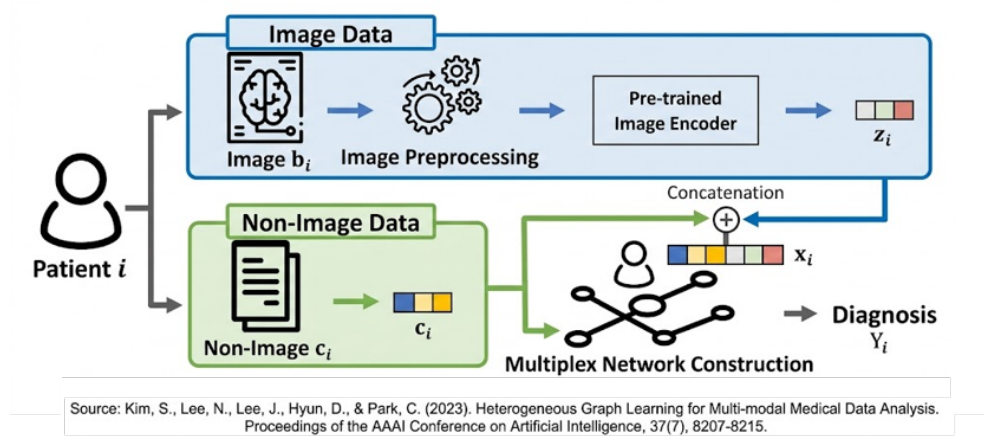


Figure 1: HetMed architecture used in the proposed model.

The final version for the proposed model is a version of the Hetmed+N4+Grayscale strategy based on the latest configurations. These configurations include updates to our earlier versions of tests using this same methodology.

5.1 LOSS FUNCTIONS

Currently, tests that directly modify the loss function were not shown to result in a substantial improvement and are not used in the final version of the proposed model.

5.2 CHANGES TO TRAINING DATA

We updated the original data described in Kim et al. (2023) from 614 to the entire patient population in the Duke-Breast data of 922 patients. We used the strategy described in Hadidchi et al. (2026) to derive tumor grade based on other columns where it was not available. This larger dataset allowed for a more balanced training split while using a test dataset with grade distributions that is still representative of the overall data. For all of the image-modality data we used the image correction strategy described in section 3 - Task Setup and Data. We used the Densenet-121-RadImageNet Mei et al. (2022) domain-specific encoder to create the initial image-modality features.

5.3 HYPERPARAMETERS AND THEIR EFFECTS

The following changes to hyperparameters resulted in improvements across all evaluation metrics:

1. Updated learning rate from 0.0001 to 0.0005.
2. Changed the original hidden dimension size of 64 to 128.
3. Changed the semi-supervised loss hyperparameter from 0.01 to 0.05 in order to place more emphasis on predicting grade labels during training.
4. Configured training to use cosine annealing with warm restarts.

6 [1 POINTS] RESULTS (1 PAGE)

Methods	Dev (Extrinsic)	
	Accuracy $\uparrow$	L_2 Error $\downarrow$
ResNet-18 (visual)	0.4667	0.5333
XGBoost (tabular only)	0.5200	0.4800
Gemini 2.5 Flash (feature only)	0.4222	0.5778
Gemini 2.5 Flash (image only)	0.2778	0.7222
MLP (tabular + visual)	0.5280	0.4720
Swin Transformer	0.5470	0.4530
MedGemma	0.5400	0.4600
Ovis2-4B	0.4500	0.5500
HetMed	0.4600	0.5400
Gemini 2.5 Flash (multimodal)	0.3667	0.6333
Gemini 3 Flash Preview (multimodal)	0.5556	0.4444
HetMed + N4 + Grayscale (v1)	0.6270	0.3730
HetMed + N4 + Grayscale (final)	0.6403	0.3597

7 [3 POINTS] ANALYSIS (2 PAGES)

HetMed+N4+Grayscale Modifications to the original version of this strategy are described in section 5. These changes reduced instability in training and resulted in better performance when evaluating embeddings generated by the model on downstream tasks. A commonality between both versions was that the consensus regularization loss contributes considerably more to the overall training when compared to other model components including metaphath-specific binary cross entropy loss and semi-supervised loss. A comparison is shown below.

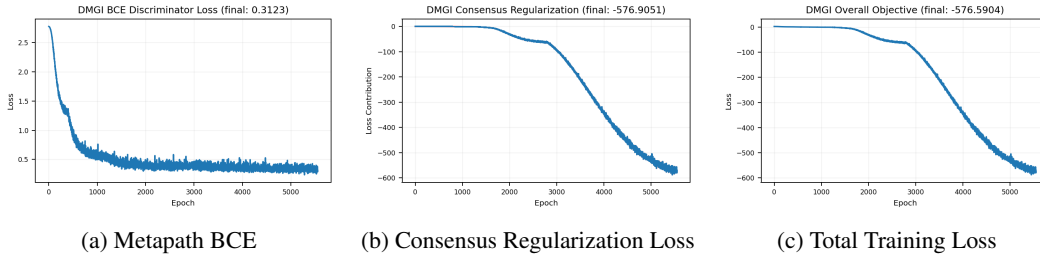


Figure 2: Loss curves highlighting the relative contribution of the consensus regularization term during training.

Graph Augmentation with HetMed The model integrates HetMed’s multiplex patient-similarity graphs into a supervised Swin+MLP classifier as a graph smoothness regularizer. From HetMed, we adopt the multiplex graph construction: four clinical relation types (receptor/staging, demographics/progression, treatment, anatomical) are each represented as binary feature vectors, normalized, and converted to patient-similarity graphs through cosine similarity thresholding. Rather than using HetMed’s full self-supervised framework (GCN encoders, mutual information maximization, consensus embedding), we add a single smoothness loss term that encourages clinically similar patients — those connected in the multiplex graphs — to have similar learned representations in the Swin+MLP embedding space. This preserves the supervised model’s simplicity and training efficiency while incorporating the patient-relationship structure that HetMed’s graphs capture.

We also implement the HetMed framework using Swin transformer as image encoder. However, both the Swin+MLP model (multimodal baseline) and the Swin+HetMed model underperform the hybrid model where the relationship matrices constructed in the HetMed framework are used to augment the learning in the original Swin pipeline.

In addition, we find out that the original graph density is imbalanced and one feature type takes over 90% of attention weight. As a result, the model is learning from only one type of relationship and much information is left unused. This is because different feature types have different numbers of features with different scales. Cosine similarity on the raw features gives higher similarity scores where features are fewer and more correlated. We expand each categorical values into its own binary column so that the number of columns reflects its actual cardinality, and apply min-max scale normalization per column to balance graph density. Furthermore, applying a universal threshold across all relationship graphs also helps decrease density differences.

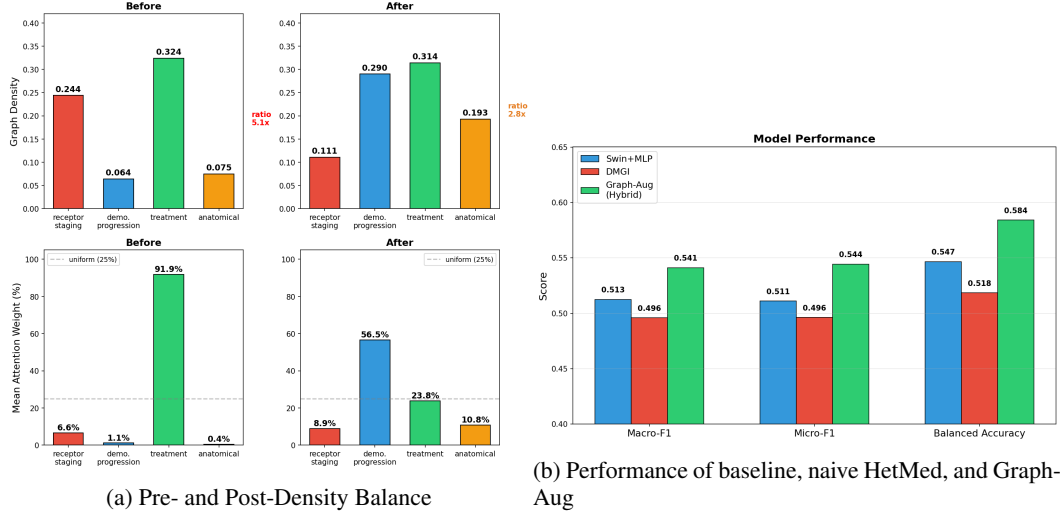


Figure 3: Model architecture decisions that help improve model performance.

7.1 INTRINSIC METRICS

Detections based on intrinsic evaluation are reported below as NMI (Normalized Mutual Information) and ARI (Adjusted Rand Index). Invalid actions are not included in this table because we did not identify any invalid actions or results that were outside of the expected bounds of generated patient representations.

Methods	Dev (Intrinsic)	
	NMI $\uparrow$	ARI $\uparrow$
XGB	0.0741	0.0325
Basic Multimodal MLP + N4	0.0298	0.0484
Basic Multimodal MLP + RGB	0.0517	0.0650
HetMed + Swin + RGB (Graph-Aug)	0.0950	0.1172
SimCLR + Grayscale	0.0066	0.0333
HetMed + N4 + RGB	-0.0099	0.0101
HetMed + N4 + Grayscale	0.0336	0.0255

Hetmed+N4+Grayscale: Intrinsic evaluations were performed before and after modifications to the final model architecture. The modality variance balance is generally unchanged after incorporating updates that include increasing the model hidden dimension size, modifications to the loss hyperparameters, and modifications to the learning rate configuration. Raw variance remains almost entirely dominated by non-image features, while after per-patient L2 normalization the split remains about 30 percent image and 70 percent non-image. While the updated version of architecture resulted in increased performance based on extrinsic results, intrinsic evaluations based on grade-specific patient cosine similarity and grade-aligned clustering did not demonstrate an improvement.

7.2 QUALITATIVE ANALYSIS AND EXAMPLES (FULL PAGE TABLES – MULTIPLE PAGES FOR MOST PROJECTS)

Visualizations for input from the four non-image feature groups described in Section 2.5 - Relevant Techniques are included below.

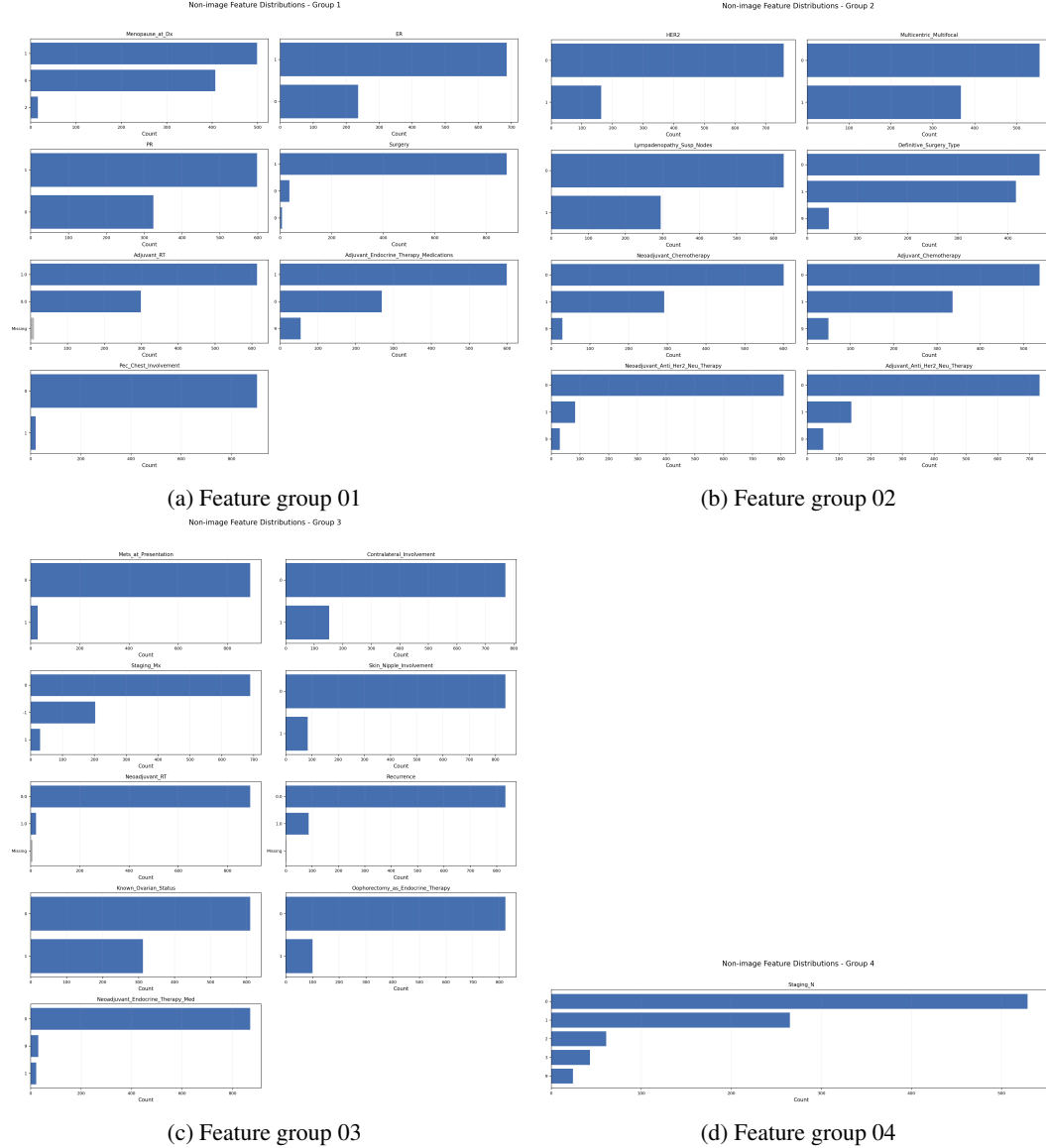


Figure 4: Non-Image Input – Feature Summaries per Feature Group.

HetMed + N4 + Grayscale: The results for the HetMed+N4+Grayscale version of our tests indicate that our strategy for N4 image correction on Grayscale images demonstrates an improvement in performance for this data. While replicating one image three times and using it as input for an image encoder is described in other studies, many of these studies describe using the subtraction image or estimating the correct series for input. Our strategy of selecting the post-contrast series of images is less common but was found to have performance comparable to other methodologies. An additional finding in this version of tests was that the selected combination for the type of image input and image encoder impacted results. Our results align with the findings of the Fridman et al. (2025) research that mentions using RGB input and encoders with an ImageNet-based architecture outperform using Grayscale images with the same encoders on this data. In our tests, specifically us-

ing the RadImageNet encoder with replication of the same grayscale image outperformed the tests that use RGB for the image modality input with the same encoder. This is potentially a result of RadImageNet Mei et al. (2022) being trained on static MRI images without encoding a temporal component. Our experiments were also designed to test differences in results based on comparing cluster-based feature selection to domain-knowledge-based feature selection methods used for creating our metapath non-image feature groups. Our results did not find that changing these feature groups had a substantial impact on performance.

Preprocessing Method	Accuracy Score	
	Cluster-Based Selection	Domain-Based Selection
N4 + Intensity Thresholding	0.64	0.63
No Image Correction	0.52	0.54

Table 2: HetMed + N4 + Grayscale Model accuracy across different image correction and feature selection strategies.

Graph-Aug: One common issue in multimodal learning is that one or few modalities dominate the learning and the model fails to capture comprehensive information. We discover that similar phenomenon occurs in multiplex graphs where one or few graphs get a majority of attention weight and dominate the learning. As shown in Figure 3, the original graph density is imbalanced and over 90% of the model’s attention is on one single feature type. It demonstrates that one feature type dominates the learning and is shown to affect model performance in Figure 5. With balanced density, the model improves classification accuracy across all three tumor grades.

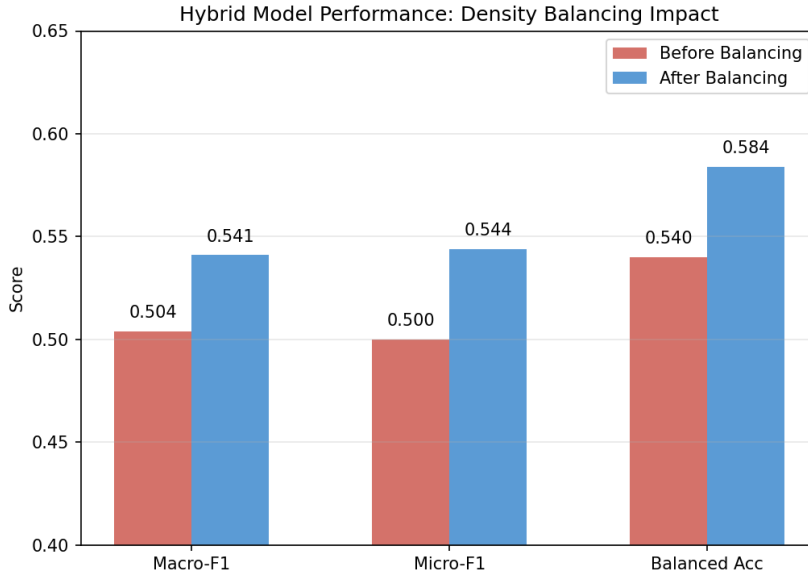


Figure 5: Performance of Graph-Aug before and after density balancing.

8 [2 POINTS] FUTURE WORK AND LIMITATIONS (1 PAGE)

HetMed + N4 + Grayscale: The current RadImageNet domain-specific encoder that is used in this strategy was trained on breast cancer ultrasound data and MRI data from knee injuries. A better variation of this would potentially be a pretrained image encoder that was trained specifically on breast cancer MRI data. Adjustments to the current configuration of this strategy are continuing to be explored along with modifications that include changing the generation of negative node permutations based on dynamically identifying importance of features. While the current performance of Macro-F1 (0.4804) and Micro-F1 (0.6403) indicates a need for improvement, our tests indicate that the changes we continue to incorporate are resulting in learning a meaningful signal as related to classifying tumor grade on highly imbalanced data. In addition, the task of predicting grade learned in the current architecture can also potentially be adapted for other tasks in the future such as Nottingham grade 2 stratification as described in Hadidchi et al. (2026).

The DMGI architecture is a representation learning framework, and as we can see from Figure 2, the consensus regularization loss term dominates the loss function. Therefore, information learned during the semi-supervised loss component of classification in our HetMed architecture may be impacted by this, hindering overall model performance. One potential future direction is to use a supervised learning framework that can be adapted to classification tasks that will better distinguish groups of embeddings for these patients.

Graph-Aug: Another variation of the HetMed model is to use the HetMed graph framework as a graph smoothness regularizer on the original Swin+MLP classifier. Even though we apply strategies to significantly improve graph density, the model still gives 50% of attention weight to one feature type, indicating that it predominantly learns from one relationship graph. A future direction is to further improve the model so that it can learn equally from all graphs and no single feature type dominates the learning. Additionally, although we resolve the issue of imbalanced prediction by achieving a much higher macro-F1 score, the overall performance of the model is only a little better than guessing. As shown in the intrinsic metrics analysis, the representation of patients is not highly discriminatory. Thus, for future work, we need to generate embeddings that capture the nuanced differences between patients with different tumor grades.

9 [1 POINTS] ETHICAL CONCERNS AND CONSIDERATIONS (UNINTENTIONAL, MALICIOUS, AND DUAL-USE)

We did not identify any ethical concerns or considerations about use for these results. This research and findings are presented with the objective of sharing results from experiments in the setting of a course project. These results are not intended for being used for any purpose outside of this course.

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A APPENDIX

You may include other additional sections here.